Learn to Bid: Near-Optimal Knapsack Bidding under Budget and CPA Constraints with Sparse Delayed Feedback

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Abstract. Real-time agentic bidders are autonomous systems that participate in programmatic advertising auctions on behalf of advertisers, making instantaneous decisions on how much to bid on ad impressions while optimizing advertiser-defined objectives such as cost-per-acquisition (CPA). These agents operate under strict budget constraints in a dynamic, multi-agent environment where auctions repeat with constantly evolving bidding dynamics. In this paper, we propose a novel budget-constrained, CPA-aware agentic bidder that learns an adaptive bidding strategy to maximize the acquisition of high-quality users while ensuring effective budget utilization under sparse and delayed feedback. At the core of our approach is a linear-time approximation algorithm that formulates bid selection under a budget constraint as a greedy knapsack optimization problem. The algorithm scales to billions of auctions per day and provides a theoretical bound on the optimality gap, yielding provable near-optimal performance. To maintain robustness in production environments, the agent incorporates real-time feedback to pace budget spend throughout the day and adapt to shifting market conditions. Without explicitly modeling competing agents, the system learns a pragmatic near-optimal bidding strategy that balances campaign performance and budget utilization.

Empirical evaluation on a large-scale demand-side platform (DSP) demonstrates that the proposed agent achieves approximately 92% of the theoretical optimum in acquisition efficiency under budget constraints. Our contributions include: (1) a scalable, near-optimal knapsack-based approximation algorithm with provable bounds; (2) a real-time adaptive bidding agent that jointly addresses budget pacing and CPA optimization under sparse and delayed feedback; and (3) extensive large-scale production validation in repeated auction environments without dominant bidding strategies.

Keywords: CPA-Aware Agentic Bidding · Real-Time Bidding · Budget-Constrained Optimization

1 Introduction

Real-time bidding (RTB) [10] platforms for mobile app acquisition operate in repeated auctions where autonomous bidders optimize advertiser objectives such as cost-per-acquisition (CPA) under strict budget constraints. These agents function in dynamic multi-agent environments with evolving auction distributions and delayed outcome feedback.

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Three structural challenges arise: (1) sparse attributed conversions despite dense upper-funnel signals, creating a mismatch between optimization objectives and observable data; (2) delayed attribution in systems such as SKAdNetwork (SKAN) [2] and MMP frameworks [1], violating dense-reward assumptions in reinforcement learning; and (3) hard performance guarantees requiring continuous CPA compliance and smooth budget pacing.

With sufficient signal, bidding follows

$$\text{bid} = \text{pCVR} \cdot \text{target-CPA}. \tag{1}$$

However, sparse high-value conversions and delayed feedback make unconstrained value bidding unstable, necessitating structural constraints.

1.1 Budget-Constrained CPA Optimization

We formulate bidding as a fractional knapsack problem: maximize expected acquisitions subject to budget B and CPA constraints. Under monotonic alignment between predicted quality and acquisition probability, the optimal allocation exhibits a greedy prefix structure governed by a scalar cutoff λ. We derive a linear-time approximation with bounded optimality gap, yielding a Budget-Constrained Bidder (BCB) that:

- Maximizes acquisitions under hard budget constraints,
- Ensures CPA feasibility via prefix optimality,
- Executes in $O(1)$ time per auction.

1.2 Adaptive Agentic Control

We embed the structural optimizer within a real-time PID-based controller that adjusts λ_t to track a smooth spend trajectory while preserving CPA guarantees. The method adapts to market dynamics without explicit competitor modeling.

1.3 Contributions

1. Linear-time CPA-constrained knapsack approximation with bounded optimality gap.
2. Structural CPA guarantees under sparse, delayed feedback.
3. Real-time adaptive cutoff control for smooth budget pacing.
4. Large-scale production validation achieving ∼92% of theoretical optimum.

This framework unifies knapsack structure with privacy-constrained RTB systems, providing a scalable and theoretically grounded solution for budget-constrained acquisition optimization.

2 Related Work

Online advertising optimization addresses budget pacing, CPA/ROI constraints, and sparse feedback in repeated auction environments. Existing approaches largely fall into two paradigms: (i) knapsack-based greedy allocation methods and (ii) optimization-based pacing frameworks grounded in dual control or reinforcement learning. While recent systems combine elements of both, most methods fundamentally rely on dense or near-instantaneous feedback—an assumption that breaks down under modern privacy-induced attribution delays.

2.1 Knapsack Auctions and Density-Based Allocation

Density-based allocation originates from classical knapsack auction formulations. Aggarwal et al. [3] study single-parameter knapsack auctions where bidders report private values v_i for publicly known sizes s_i. The fractional knapsack solution sorts bidders by density $\rho_i = v_i/s_i$ and selects a prefix. Although this greedy rule maximizes social welfare in the fractional relaxation, it is not inherently truthful. The authors therefore design deterministic and randomized truthful mechanisms that achieve constant-factor approximations (e.g., 1/2 deterministically). While foundational, this line of work does not incorporate advertiser-side budget pacing or CPA/ROI feasibility constraints.

Google Research [8] operationalized the fractional knapsack intuition in sponsored search by ranking queries according to ROI density and throttling participation probabilistically after the selected prefix to satisfy budget feasibility in expectation. Specifically, participation probability is set to $p = B/\hat{S}$, where B is total budget and $\hat{S}$ is predicted spend of the greedy prefix. This approach preserves ranking optimality but enforces constraints only in expectation.

Khezr et al. [9] extend knapsack auctions to strategic settings under discriminatory (DSA), generalized second-price (GSP), and uniform-price (UP) auctions, showing trade-offs between revenue and truthfulness across formats. However, their analysis abstracts away from query-level sparsity and delayed conversion feedback.

Hao et al. [5] study dynamic multi-channel budget allocation via bilevel optimization. The upper level greedily selects users according to CPR density $\rho_u = V_u/S_u$ (predicted lifetime value V_u over expected spend S_u), while the lower level trains per-user bidding policies using reinforcement learning (e.g., DQN, DDPG, PPO) with reward $r_t = v_t - \tau_t c_t$. Although deployed at scale with significant ROI gains, the framework relies on user-level state transitions and relatively dense feedback, limiting robustness under sparse and delayed attribution.

In contrast to prior density-based systems that rely on expectation-level feasibility or dense feedback, our work establishes that under monotonic calibration, CPA-constrained optima retain a strict greedy prefix structure. This allows joint hard feasibility of both budget and CPA constraints without probabilistic throttling.

2.2 Optimization-Based Pacing and Dual Control

An alternative paradigm formulates bidding as a constrained optimization problem solved via dual updates or reinforcement learning.

For example, [6] formulate bidding as a constrained linear program and derive a parametric bid function $b_i = \sum_j w_j \, \mathrm{score}_{i,j}$, where weights are dynamically learned using Deep Deterministic Policy Gradient (DDPG). Such approaches adapt to market fluctuations but depend on sufficiently dense feedback to stabilize policy learning.

Feng et al. [4] provide theoretical guarantees for RoS-constrained bidding using a primal-dual framework with Online Mirror Descent (OMD), achieving near-optimal regret under i.i.d. request streams. While dual methods enforce constraints in expectation, effective updates require near-immediate reward observation.

These optimization-based approaches assume reward signals arrive quickly enough to update dual variables or policies with low variance. Under privacy-centric attribution systems such as SKAdNetwork or delayed MMP reporting, feedback latency introduces substantial instability, weakening regret guarantees and complicating constraint enforcement.

2.3 Online Learning under Delayed Feedback

The theoretical study of delayed feedback in online learning was formalized by [7], who proposed black-box meta-algorithms (e.g., BOLD) that transform standard learners into delay-robust variants. They show that delay increases regret additively in stochastic settings and multiplicatively in adversarial environments. However, this literature focuses on unconstrained regret minimization and does not address joint budget and CPA feasibility.

Rather than absorbing delay through regret-compensating learner instances, we exploit a structural property of the fractional knapsack problem: prefix optimality under monotonic density ordering. This reduces the multidimensional constrained bidding problem to a one-dimensional control rule

$$b_i = \lambda_t q_i,$$

where λ_t is a dynamically regulated marginal cutoff. By separating structural allocation from pacing control, we ensure hard budget and CPA guarantees even when attribution is sparse and delayed.

2.4 Positioning of Our Work

Unlike prior work that (i) enforces constraints only in expectation, (ii) relies on dense reward signals for stable dual updates, or (iii) treats delay through regret inflation, we establish joint hard feasibility via prefix monotonicity under realistic sparsity and attribution delay. The resulting agent reduces bidding to an $O(1)$ scalar control update without reinforcement learning or mirror-descent overhead, while maintaining provable near-optimality relative to the fractional knapsack benchmark.

3 Budget-Constrained Bidder

Pricing is the core challenge of a Demand-Side Platform (DSP): determining per-request bids that maximize campaign objectives while respecting budget constraints. In first-price Real-Time Bidding (RTB), bids must balance winning probability and cost efficiency. Under a fixed budget B, naïve constant bidding,

$$\mathrm{bid}_i = b',$$

maximizes impressions subject to

$$\max \sum_i X_i \quad \text{s.t.} \quad X_i = \mathbb{I}[b' > cp_i], \quad \sum_i b' X_i \leq B,$$

Constant bidders fail to distinguish between high- and low-quality users, treating all requests equally and often winning low-value impressions. However, the objective of a DSP is to maximize total acquired quality under budget constraints, not merely impression volume. To address this, linear budget-constrained bidders (BCBs) bid proportionally to user quality as

$$\mathrm{bid}_i = \lambda q_i,$$

where q_i is a quality score and λ is a scalar multiplier. The optimization becomes

$$\max \sum_i q_i X_i \quad \text{s.t.} \quad b_i = \lambda q_i, \quad \sum_i b_i X_i \leq B, \quad X_i = \mathbb{I}[b_i > cp_i].$$

Actual challenge is in determining the right λ value that maximizes utility within the given budget, considering the market competition. Our goal can be mathematically expressed as

$$\lambda^\star = \arg\max_\lambda \sum_i q_i X_i \quad \text{s.t. budget feasibility.}$$

Knapsack Interpretation. Viewing budget as knapsack capacity, clearing price cp_i as weight, and quality q_i as value, the relaxed fractional solution sorts by density q_i/cp_i and selects a prefix. Let j denote the last selected request. Setting

$$\lambda = \frac{cp_j}{q_j}$$

This solution is optimal for the fractional knapsack, where the final request may be partially selected. Since requests are indivisible, we discard the fractional request r_j with fraction $f \in (0, 1)$, yielding a feasible solution with budget $(B - f \cdot b_j)$. Because $b_j \ll B$, the optimality gap is negligible and bounded by b_j.

The key challenge is determining λ such that all requests in the greedy knapsack are won while others are excluded. This is achieved by choosing λ so that the bid of the marginal request equals its clearing price, ensuring selection of the optimal prefix.

Proposition 1. *Choosing $\lambda = cp_j/q_j$ (for the last prefix element j) wins exactly the greedy knapsack set.*

Proof. For i before j,

$$\frac{q_i}{cp_i} > \frac{q_j}{cp_j} \implies \frac{cp_i}{q_i} < \frac{cp_j}{q_j} = \lambda,$$

so $b_i = \lambda q_i > cp_i$.

λ can be efficiently computed using binary search over all the auction requests. This approach is enabled by the monotonic relationship between λ and expected spend: increasing λ monotonically increases bid values, which in turn increases the probability of winning auctions and therefore cumulative spend. Consequently, the budget-feasible solution lies on a one-dimensional monotonic curve, making binary search both efficient and theoretically justified.

3.1 Optimizing for CPA

Assume acquisition probability satisfies $p_i = f(q_i)$ with f monotone increasing. Then maximizing $\sum_i q_i X_i$ under budget is equivalent to maximizing expected acquisitions $\sum_i p_i X_i$. Explicit estimation of f is unnecessary for acquisition maximization **as long as the quality score is monotonically aligned with acquisition probability**.

3.2 CPA-Constrained Extension

While the budget-constrained formulation maximizes acquisitions, it does not directly enforce a target cost-per-acquisition (CPA). Incorporating CPA constraints requires estimation of the mapping function f, which can be learned statistically as campaign data accumulates.

Once f is estimated, the optimization can be extended to incorporate CPA-aware decision-making:

- If the target CPA is achievable within the available budget, the system optimizes toward minimizing CPA while maximizing acquisitions.
- If the target CPA is infeasible given the budget and auction environment, the system instead determines the maximum achievable spend that satisfies the CPA constraint.

Extended formulation of optimization problem is as follows

$$\lambda^\star = \arg\max_\lambda \sum_i q_i X_i$$

subject to

$$\sum_i b_i X_i \leq B, \quad \sum_i b_i X_i \leq \mathrm{CPA}_{\max} \sum_i p_i X_i.$$

Proposition 2 (Greedy knapsack yields best achievable CPA under monotone calibration). *Assume the true acquisition probability is a monotone function of the quality score, i.e.,*

$$p_i = f(q_i), \qquad \textit{with } f \textit{ monotone increasing in } q.$$

Consider two requests i and $i{+}1$ such that the greedy ordering holds:

$$\frac{q_i}{cp_i} > \frac{q_{i+1}}{cp_{i+1}}.$$

Then the corresponding expected CPAs satisfy

$$\mathrm{CPA}_i < \mathrm{CPA}_{i+1},$$

where

$$\mathrm{CPA}_k \triangleq \frac{cp_k}{p_k}.$$

Consequently, the greedy knapsack ordering by $\frac{q}{cp}$ packs requests in non-decreasing expected CPA, and thus achieves the best possible (minimum) average CPA among all feasible packs of the same cardinality (equivalently, for any prefix length).

Proof. Greedy ordering implies

$$\frac{q_i}{cp_i} > \frac{q_{i+1}}{cp_{i+1}} \iff \frac{cp_i}{q_i} < \frac{cp_{i+1}}{q_{i+1}}.$$

Monotonicity gives $p_i \geq p_{i+1}$, and ratio monotonicity yields

$$\frac{cp_i}{p_i} < \frac{cp_{i+1}}{p_{i+1}},$$

so expected CPA increases along the prefix.

Greedy ordering by $\frac{q}{cp}$ is consistent with ordering by increasing expected CPA. Thus, greedy knapsack remains CPA-optimal under monotone calibration.

Remark 1 (Calibration not required). The result does not require calibrated scores. It suffices that acquisition probability is an order-preserving transform $p = f(q)$ with monotone f and ratio-monotonicity $\frac{f(q)}{q}$ non-decreasing. Thus, even with uncalibrated but monotone scores, greedy packing by $\frac{q}{cp}$ maintains increasing expected CPA and achieves the best feasible CPA for the packed set.

3.3 Dual Interpretation of λ

The acquisition-maximization problem, with c_i as cost and v_i as value, is

$$\max_{x \in \{0,1\}^N} \sum_i v_i X_i \quad \text{s.t.} \quad \sum_i c_i X_i \leq B.$$

Its Lagrangian is

$$\mathcal{L}(X, \lambda) = \sum_i (v_i - \lambda c_i) X_i + \lambda B,$$

yielding

$$X_i^\star = \mathbb{I}[v_i - \lambda c_i > 0].$$

Thus, λ is the *shadow price of budget*, capturing the marginal value of spend. The Budget-Constrained Bidder (BCB) operationalizes this via a single scalar λ to scale bids (e.g., $\text{bid}_i = \lambda q_i$), tuned so realized spend matches budget. Since expected spend is monotone in λ, tuning is efficient (e.g., binary search or closed-loop control).

3.4 Bootstrapping Strategy

New advertisers begin as purely budget-constrained bidders to collect data and estimate f. As calibration improves, the system transitions to CPA-aware optimization, either minimizing CPA under budget (if feasible) or restricting spend to satisfy the CPA constraint.

4 Real-Time Adaptation

Auction dynamics are non-stationary; a static λ can cause over or under-delivery. We therefore implement a closed-loop control system that adjusts λ intra-day based on spend feedback while maintaining stability bounds.

4.1 Spend Allocation

For campaign i with budget b_i, discretize day T into slots $t \in \{1, \ldots, 1440\}$. Let cumulative spend be $s_{i,t}$. Using forecasts, we define ideal cumulative spend:

$$\text{ideal\_spend}_{i,t} = \frac{f_{i,t}}{O_{i,T}} b_i, \tag{2}$$

and bids are $q_i \cdot \lambda$.

We use two pacing strategies:

– **Opportunity-based:** Allocate spend proportional to expected volume.
– **Quality-based:** Allocate proportional to predicted quality mass $\sum q_i$.

Quality-based pacing is superior in practice, shifting spend toward high-value periods and improving CPA.

4.2 Control Update

Define pacing error:

$$e_t = \text{ideal\_spend}_{i,t} - s_{i,t}. \tag{3}$$

We update λ via PID control:

$$\Delta\lambda_t = K_p e_t + K_i \sum_{\tau=1}^{t} e_\tau + K_d(e_t - e_{t-1}). \tag{4}$$

– K_p: immediate correction; K_i: removes persistent bias; K_d: stabilizes rapid shifts.

Parameters are campaign-specific, tuned via Monte Carlo simulation with Dirichlet allocation. λ is bounded to prevent bid collapse or runaway bids.

5 Experiments

In this section, we delve into the details of the experimental setups in detail and present the results from our empirical evaluation of the proposed algorithm with CPA-aware extension and real time λ adaption. Our goal of our evaluation is multi-fold:

1. Quantification of the optimality gap relative to a clairvoyant oracle.
2. Validation of budget adherence under real world auction dynamics.
3. Demonstration of enforcement of strict CPA enforcement.

All the experiments were conducted on live traffic in our production environment under first-price auctions.

5.1 Experiment 1: Comparison against the oracle

This experiment measures how close the λ-controlled BCB operates in online setting, relative to the clairvoyant oracle that retrospectively computes the best possible λ^* using full-day information, i.e. the oracle knows the full-day clearing price in advance. This oracle presents the theoretical upper bound achievable under:

1. Budget constraint
2. Clearing price realization
3. Quality scores

The only difference is that the oracle selects λ^* using perfect foresight, whereas our system must estimate and adapt λ online using real-time feedback.

Experimental Setup

We selected five heterogeneous advertisers spanning different categories, budget ranges, quality score distributions, and targeting constraints. For each advertiser, we ran the experiment over 7 days to ascertain the effectiveness while ensuring the bidder is exposed to diverse temporal patterns and auction dynamics. For each advertiser-day pair, we logged all the auction opportunities, its associated clearing price and predicted quality score. Using this data, we then computed the oracle λ^*, that is the value of λ that would have maximized the total quality under realized clearing prices. We then compared a range of performance metrics between this oracle benchmark and the λ_t produced by the online system.

The optimality gap is defined as:

$$\text{Optimality Gap (\%)} = \frac{\text{Oracle Quality} - \text{Online Quality}}{\text{Oracle Quality}} \times 100$$

Results

Across all advertisers, the average optimality gap was observed to be 8.06%. The table below summarizes the performance per advertiser across a 7-day period.

Table 1. Performance comparison across advertisers: 7-days evaluation.

Advertiser	Budget ($)	Average Opportunities	Average Wins	Win Rate (%)	Wins over Organic Users	Budget Utilization (%)	Oracle's Cumulative Quality	Online's Cumulative Quality	Optimality Gap (%)
A1	2,000	8,214,739	382,514	4.6552	1,246	99.184	4,182.73	3,847.91	8.001
A2	3,500	11,906,284	642,973	5.4017	2,684	98.763	7,964.51	7,319.38	8.101
A3	6,000	17,284,915	865,421	5.0068	3,742	99.327	13,548.84	12,437.09	8.207
A4	8,000	22,741,608	1,118,392	4.9175	4,906	98.541	18,276.39	16,811.82	8.012
A5	9,200	27,968,442	1,603,714	5.7338	6,811	98.904	24,915.63	22,931.56	7.999
Avg.	—	—	—	—	—	**98.944**	—	—	**8.064**

Interpretation of the results

1. Tight optimality gap: Despite not knowing clearing prices in advance, the online system operates consistently within $\tilde{8}$% of the oracle's upper bound. This confirms that the fractional knapsack is practically tight and real-time adaptation of λ compensates effectively for evolving auction environment.
2. Robustness across budget ranges: Budget utilization has remained consistently above 98%. This effectively indicates that the optimality gap remains stable across small and large campaign budgets.
3. Scalability: The algorithm can scale to billions of auction requests per day with a O(1) inference per request, requiring only a scalar update to λ.

5.2 Experiment 2: CPA constrained convergence

While the previous experiment validates the optimality of the budget-constrained bidder, real-world campaigns typically operate under strict CPA constraints. In this experiment, we evaluate how the CPA-constrained BCB behaves under conditions where the target CPA is significantly lower than what is achievable at the given budget.

Campaign selection

We selected a live campaign for this experiment that had been active on budget-constrained bidder for a sufficient period of time to ensure stable operating conditions. The characteristics of the campaign and the expectation of the advertiser is summarized below:

1. Daily budget: ≈ $2000
2. Target CPA: ≈ $11
3. CPA observed, before intervention: ≈ $32
4. Attribution delay: ≈ $24h$

During the observation period of 7 days, the campaign operated under budget-constrained bidding where-in the target CPA was not explicitly enforced.

Performance summary

The table below illustrates campaign performance prior to on-boarding, where bidding was budget-constrained and spend-optimized, versus post on-boarding performance under a CPA-constrained bidder with strict CPA control.

Table 2. Performance summary across optimization phases.

Phase	Avg Daily Spend ($)	Avg CPA ($)	Avg Daily Acquisitions	Budget Utilization (%)
Week 1 (Budget Only)	2,000	32.4	62	100
Week 2 (Transition)	1,200	16.7	72	60
Week 3 (Converged)	980	11.8	83	49
Week 4 (Stabilized)	1,020	10.9	94	51

Observed behavior

Week 1: Pre-CPA Constraint

- Daily spend: ≈ $2000
- Average CPA: ≈ $32
- Strong budget adherence.
- Severe performance violation.

After 7 days, the campaign was transitioned to CPA-constrained BCB.

Week 2: Constraint Activation

- λ adjusted downward due to CPA constraint.
- Lower-density or high CPA requests are pruned.
- Spend drops sharply.
- CPA rapidly improves.

Week 3 & 4: Equilibrium

- Spend stabilizes to $\approx \$1000$.
- CPA converges to $\approx \$11$.

The above observed behavior confirms a key structural property: If the target CPA is infeasible at full budget, the algorithm starts to honor the give input CPA constraint and converges to a maximum feasible spend that satisfies the constraint.

6 Key Takeaways

- **Near-oracle efficiency:** Achieves ∼92% of oracle performance under budget constraints.
- **Structural CPA enforcement:** Constraints satisfied deterministically, not in expectation.
- **Automatic feasibility adjustment:** Spend reduces when CPA binds.
- **Scalable deployment:** Practical at billion-scale auctions.

Overall, reducing bidding to a one-dimensional λ optimization via prefix optimality yields a theoretically sound and operationally robust system for large-scale repeated auctions.

7 Future Work

Our current framework computes a campaign-level λ that is locally optimal under each campaign's budget constraint. While near-optimal in isolation, this approach does not account for cross-campaign interactions within the DSP.

In practice, campaigns frequently compete for overlapping inventory due to shared targeting (e.g., geography, device, cohorts, contextual or demographic filters). When independently optimal bidders enter the same auction, they effectively compete against one another. This can lead to:

1. Internal bid inflation
2. Opportunity cannibalization
3. Reduced overall DSP welfare

7.1 Toward a Globally Optimal Knapsack Formulation

From the DSP perspective, the objective is to maximize total platform welfare across campaigns under budget constraints—not merely individual campaign utility. We therefore propose extending the framework to a *Global Stochastic Knapsack (GSK)* formulation, where:

1. Campaigns are optimized jointly,
2. Allocation accounts for cross-campaign value,
3. The objective maximizes overall DSP welfare.

Each auction opportunity becomes a multi-valued item whose value depends on the winning campaign. The platform allocates impressions to the campaign generating highest marginal welfare while respecting all budget constraints.

8 Conclusion

We introduced a budget-constrained, objective-aware agentic bidder for large-scale programmatic auctions. By formulating bidding as a knapsack-style problem, we developed a linear-time approximation algorithm that scales to billions of auctions per day with provable optimality guarantees. Deployed in production at a large scale DSP, the system adapts in real time to evolving auction dynamics and achieves $\approx 92\%$ of the theoretical optimum in acquisition efficiency.
Overall, this work shows that well-designed approximation methods combined with real-time adaptation can deliver both scalability and strong practical performance in modern advertising systems.

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